

ACAD

Global AI Assistant & Sales Platform

Product Requirements Document (PRD)
Hoduction Technologies

Standalone: acad.hoduction.tech

Integrated with: hoduction.tech, acados.app, and future Hoduction product websites

Deployment: Independent Vercel application + centralized AI API

CORE PRINCIPLE

One AI brain. Multiple interfaces. One centralized knowledge, tool, sales and conversation system.

1. Product Vision

Acad is the centralized AI intelligence layer for Hoduction Technologies. It must operate as a complete standalone AI application while also being embeddable into Hoduction and product websites.

Acad combines four major roles: **Knowledge Assistant, Product Consultant, Demo Guide, and Sales Qualification Assistant.**

It must answer company/product questions, understand visitor context, explain products and modules, retrieve official knowledge, show relevant simulations, qualify leads, collect enquiries, send enquiry notifications through Resend, and maintain consistent behavior across every interface.

2. Non-Negotiable Architecture

Do NOT build separate chatbot implementations for each website. Every interface must use the same centralized Acad Core/API.

```
ACAD CORE
LLM + RAG + Memory + Tools
Knowledge + Sales + Analytics
|
Central API
/ | \
acad.hoduction.tech hoduction.tech acados.app
Standalone App Widget Widget
|
Future Websites
```

The standalone app, Hoduction website and AcadOS website may have different visual context, but their AI behavior and knowledge must originate from the same core.

3. Deployment & URLs

- **acad.hoduction.tech** — standalone full Acad application.
- **acad API** — centralized backend/API serving every interface. Exact production API hostname may be selected during implementation.
- **widget.js** — reusable embeddable widget served from the Acad deployment.
- **hoduction.tech** — company website consuming Acad through the widget/API.
- **acados.app** — product website consuming the same Acad core.

4. Global Consistency Requirement

The same user question asked through acad.hoduction.tech, a widget on hoduction.tech, or a widget on acados.app must be answered by the same central AI system.

Only site/product/page context should change the response where relevant. There must never be separate prompts, RAG systems, model logic, lead systems, or tool implementations that cause one deployment to behave differently from another.

5. Context-Aware Chat

Every embedded instance must provide context to the central API, including site, product and current page.

```
{
  "site": "acados",
  "product": "AcadOS",
```

```
"page": "/testmaker"  
}
```

This allows Acad to understand what the visitor is currently viewing and answer in a context-aware manner.

6. Knowledge & RAG

Acad must use a proper RAG architecture rather than relying on a giant system prompt.

```
Knowledge Sources  
↓  
Content ingestion  
↓  
Extraction + chunking  
↓  
Embeddings  
↓  
Supabase PostgreSQL + pgvector  
↓  
Semantic retrieval  
↓  
Relevant context  
↓  
LLM response
```

Knowledge sources

- Hoducation company information
- Services and process
- AcadOS overview and modules
- Official product documentation
- Website pages
- FAQs
- PDF/DOCX/Markdown documents
- Product and module data
- Interactive simulation metadata

7. Knowledge Source Priority

When answering company/product questions, use this priority:

- Structured Product Data
- Official Knowledge Base
- Official Website Content
- Official Documentation
- Simulation Metadata
- Conversation Context
- General LLM Knowledge

Acad must never invent prices, features, integrations, customers, timelines, guarantees or other business claims. If verified information is unavailable, it must say so.

8. Supabase Data Model

Use Supabase PostgreSQL as the central data layer and pgvector for semantic retrieval.

```
organizations
products
modules
knowledge_sources
knowledge_documents
knowledge_chunks
knowledge_embeddings
faqs
website_pages
simulations
simulation_steps
conversations
messages
leads
enquiries
tool_calls
analytics_events
sites
```

9. Website / Product Sync

Website and product information should be ingestible into the central knowledge system. Do not require model retraining for routine content changes.

```
Website/CMS
↓
Content ingestion / webhook
↓
Extract → Chunk → Embed
↓
Central Knowledge Base
↓
Acad immediately uses updated knowledge
```

n8n may be used later as an automation layer for website/CMS webhooks and knowledge ingestion.

10. Embeddable Widget

Create a reusable widget that can be installed on any Hoduction property.

```
src="https://acad.hoduction.tech/widget.js"
data-site="hoduction"
data-product="company">

src="https://acad.hoduction.tech/widget.js"
data-site="acados"
data-product="acados">
```

The widget must not contain the central AI business logic. It should communicate with the Acad API.

11. Acad Capabilities

- Answer company questions.
- Answer product and module questions.
- Explain use cases and workflows.
- Answer technical/pre-sales questions using verified knowledge.

- Understand page/product context.
- Maintain conversation context.
- Recommend relevant products/modules.
- Show relevant interactive simulations.
- Qualify buying intent.
- Collect enquiries.
- Send enquiry information through Resend.
- Generate useful conversation summaries for sales.

12. Sales Qualification

Acad should act like a product consultant, not an aggressive sales bot.

```

Visitor
↓
Understand requirement
↓
Answer questions
↓
Recommend relevant solution
↓
Show simulation/demo
↓
Handle objections/questions
↓
Identify buying intent
↓
Offer demo/enquiry
↓
Capture lead
↓
Notify sales

```

Lead stages may include: **EXPLORING, QUALIFIED, DEMO_REQUESTED, ENQUIRY_SUBMITTED**. These are transparent conversation states, not arbitrary judgments.

13. Lead / Enquiry Data

```

name
email
phone
organization
organization_type
number_of_students
number_of_branches
current_system
requirements
interested_product
interested_module
timeline (optional)
budget_range (optional)
conversation_summary
source_site
source_page
timestamp

```

Do not ask for every field at once. Collect information naturally as the conversation develops.

14. Enquiry Form

Provide a native enquiry form with at least:

- Full Name *
- Organization *
- Email *
- Phone *
- Requirement *
- Product / Service
- Number of Students
- Number of Branches
- Message
- Preferred Demo Date (optional)

15. Resend Integration

Use the existing Resend API configuration available in the Antigravity environment. API credentials must remain server-side.

When an enquiry is submitted, store it and send a structured email to the configured Hoduction sales/contact email.

```
NEW ACAD ENQUIRY
```

```
Name:  
Organization:  
Email:  
Phone:
```

```
Interested Product:  
Interested Module:
```

```
Students:  
Branches:
```

```
Requirement:
```

```
Conversation Summary:
```

```
Source Website:  
Source Page:
```

```
Timestamp:
```

16. AcadOS Simulation Integration

Acad must understand the interactive simulations/demos already available on acados.app and recommend them contextually during conversations.

If a user asks about a feature that has a relevant simulation, Acad should answer the question and naturally offer the simulation instead of merely describing it.

```
simulations  
id  
product  
module
```

```
title
description
url
type
status
metadata
```

Expected examples include TestMaker, CBT, OMR, ERP and CRM simulations where those demos exist and are registered.

17. Simulation Decision Logic

```
IF user asks about a feature
AND relevant simulation exists
THEN answer the question
AND offer the simulation.

IF user expresses product interest
THEN explain the relevant solution
AND show a useful simulation where available.

IF user shows genuine buying intent
THEN answer remaining questions
AND offer demo/enquiry.

IF enquiry is submitted
THEN save lead
AND send Resend notification
AND confirm submission to the user.
```

18. AI Tools / Function Calling

Implement a centralized tool layer. Suggested tools:

- search_knowledge()
- get_product()
- get_module()
- find_simulation()
- capture_lead()
- submit_enquiry()
- send_enquiry_email()
- book_demo() — future
- check_demo_availability() — future

All tools must be centrally implemented and permission-controlled.

19. Conversation Memory

Within a session, Acad should remember relevant details such as the visitor's requirement, organization size, products/modules discussed and simulations already viewed.

Store conversations using secure visitor/session identifiers. Do not expose private conversation data merely because a visitor moves between websites.

20. Standalone UI

acad.hoducation.tech should be a polished, responsive full-screen AI experience with:

- Acad branding
- Welcome message
- Suggested questions/actions
- Streaming responses
- Markdown support
- Conversation history
- Simulation cards
- Enquiry form
- Mobile responsive design

The standalone UI is an independent interface only; it must still use the same central AI API.

21. Admin Dashboard

Create an internal dashboard for:

- Conversation analytics
- Leads and enquiries
- Knowledge sources/documents
- Products and modules
- Simulation registry
- Site configuration
- Tool logs
- Top questions
- Top products/modules
- Conversion funnel

22. Multi-Site Configuration

```
sites
site_id
name
domain
product
enabled
theme
welcome_message
allowed_knowledge
```

A new Hoducation product website should be able to integrate Acad by registering the site/product context and installing the widget, without building a new chatbot.

23. Security

- Server-side LLM API calls

- Server-side Resend credentials
- API authentication
- Site/API keys
- CORS restrictions
- Rate limiting
- Input validation
- Admin authorization
- Tool permission controls
- Audit/logging
- No secrets in frontend code

24. Failure Handling

If the AI provider or knowledge system is unavailable, provide a graceful response and allow the visitor to submit an enquiry. Never fabricate an answer to hide an outage or knowledge gap.

25. Observability

Track at minimum:

- conversation_id
- message_id
- model
- latency
- token usage
- tool calls
- retrieval results
- errors
- site
- page
- product

26. Recommended Tech Stack

Layer	Technology
Standalone frontend	Next.js + TypeScript
Central API	Node.js / TypeScript
Database	Supabase PostgreSQL
Vector search	pgvector
AI	Primary LLM provider, abstracted behind AI service

Automation	n8n
Email	Resend
Deployment	Vercel
Source control	GitHub

27. Recommended Monorepo Structure

```

acad/
├── apps/
│   ├── acad-web/ # acad.hoducation.tech
│   ├── acad-api/ # central AI backend
│   ├── acad-widget/ # embeddable widget
│   └── acad-admin/ # internal dashboard
├── packages/
│   ├── ai-core/
│   ├── rag/
│   ├── knowledge/
│   ├── tools/
│   ├── simulations/
│   ├── leads/
│   ├── email/
│   ├── database/
│   └── shared-types/
├── docs/
│   ├── PRD.md
│   ├── ARCHITECTURE.md
│   ├── AI_SYSTEM.md
│   ├── KNOWLEDGE.md
│   ├── TOOLS.md
│   ├── SIMULATIONS.md
│   ├── SECURITY.md
│   ├── DEPLOYMENT.md
│   └── README.md

```

28. Environment Variables

```

AI_PROVIDER_API_KEY=
SUPABASE_URL=
SUPABASE_SERVICE_ROLE_KEY=
RESEND_API_KEY=
ACAD_API_SECRET=

```

Never hard-code secrets or expose them to browser/client bundles.

29. Development Phases

V1 — ACAD CORE

Standalone chat, central API, LLM, Supabase, conversation history, streaming, basic knowledge.

V2 — RAG

Website ingestion, documents, embeddings, pgvector, semantic retrieval, product/module knowledge.

V3 — GLOBAL WIDGET

Reusable widget, site/product/page context, Hoducation integration, AcadOS integration.

V4 — SALES

Enquiry form, lead capture, Resend, qualification states, conversation summaries.

V5 — SIMULATIONS

Simulation registry, contextual demo recommendations, TestMaker/CBT/OMR/ERP/CRM demo support.

V6 — ADMIN

Knowledge management, simulations, conversations, leads, analytics and site configuration.

30. Acceptance Criteria

- Standalone Acad deployment works at acad.hoduction.tech.
- Central API serves standalone and embedded experiences.
- hoduction.tech widget works.
- acados.app widget works.
- All interfaces use the same AI core, knowledge, tools and lead system.
- Site/product/page context reaches the central API.
- RAG returns official, relevant knowledge.
- Acad does not invent unsupported company/product facts.
- Enquiry form works and saves leads.
- Resend notification works with server-side credentials.
- Conversation summary is included in sales enquiry.
- Simulation registry works and relevant simulations can be recommended.
- TestMaker/CBT/OMR/ERP/CRM simulations can be registered where available.
- Conversation context is maintained within sessions.
- Admin dashboard provides core knowledge, lead, simulation and analytics management.
- Secrets are never exposed client-side.
- The same question produces consistent knowledge-based behavior across all interfaces, with only relevant site/product context changing the response.

31. Master Instruction for Antigravity

Build ACAD as an independent, production-ready AI platform for Hoduction Technologies.

Acad must be deployed independently on **acad.hoduction.tech** as its own Vercel application. It must NOT be implemented as separate chatbot logic inside Hoduction or AcadOS websites.

Build a centralized AI Core/API containing LLM orchestration, RAG, knowledge retrieval, conversation context, tool calling, lead qualification, enquiry handling, simulation registry and analytics.

Build three consumption layers: (1) standalone Acad web application, (2) reusable embeddable Acad widget, and (3) central API used by both.

hoduction.tech, acados.app and all future Hoduction product websites must use the exact same Acad Core/API. The embedded widget must send website context including site, product and current page URL.

Use Supabase/PostgreSQL + pgvector for the central data and RAG layer. Use the existing Resend configuration available in the Antigravity environment for enquiry notifications. Keep all credentials server-side.

Inspect the existing AcadOS website and its modules/simulations before implementation. Register available interactive simulations from acados.app so Acad can recommend them contextually.

Build the system in phases. Before major implementation, create PRD.md, ARCHITECTURE.md, AI_SYSTEM.md, KNOWLEDGE.md, TOOLS.md, SIMULATIONS.md, SECURITY.md and DEPLOYMENT.md.

Do not overwrite existing AcadOS functionality. Acad must integrate through a reusable widget/API architecture. The final system must behave as one global chatbot across the entire Hoducation ecosystem.

END OF PRD